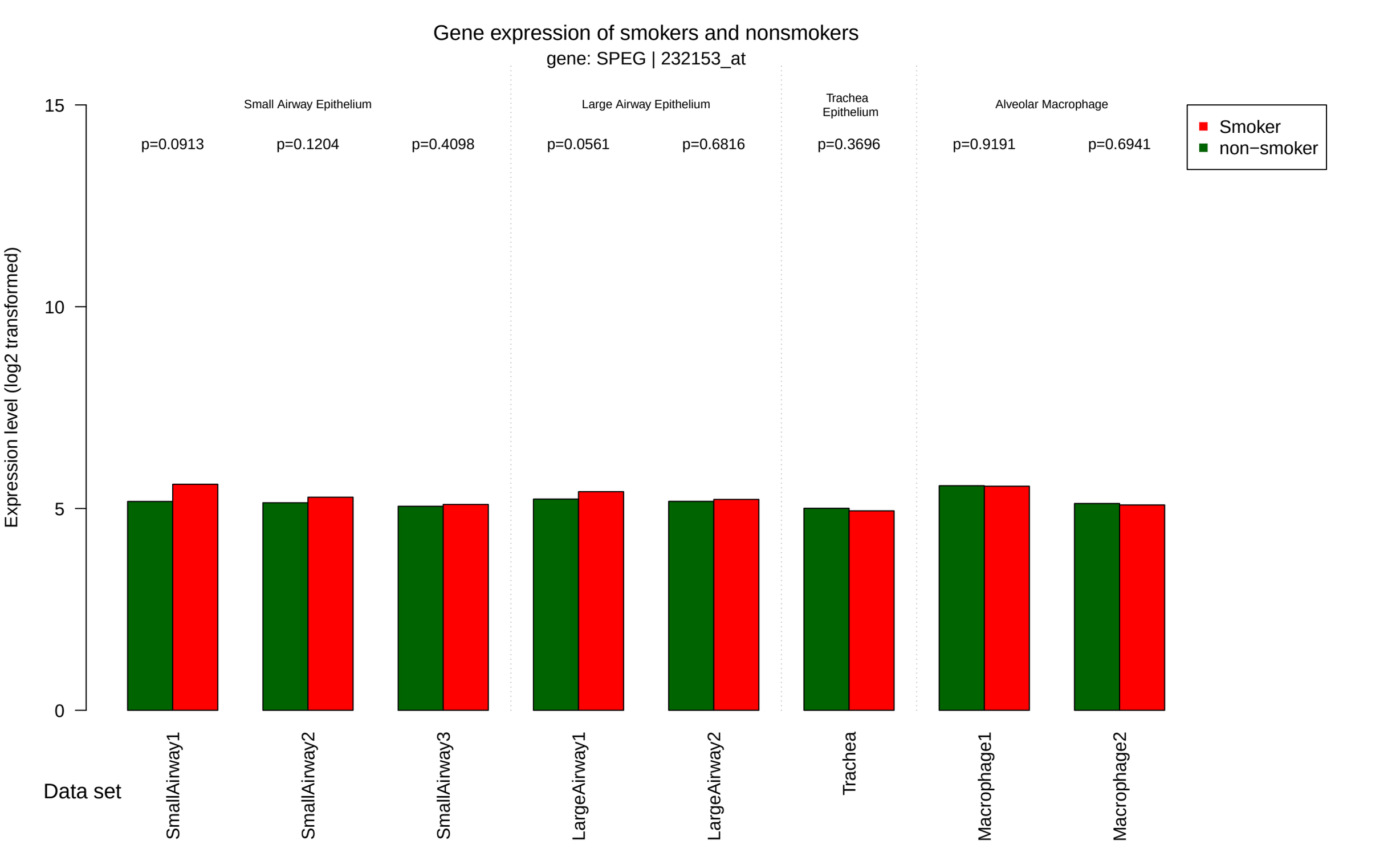
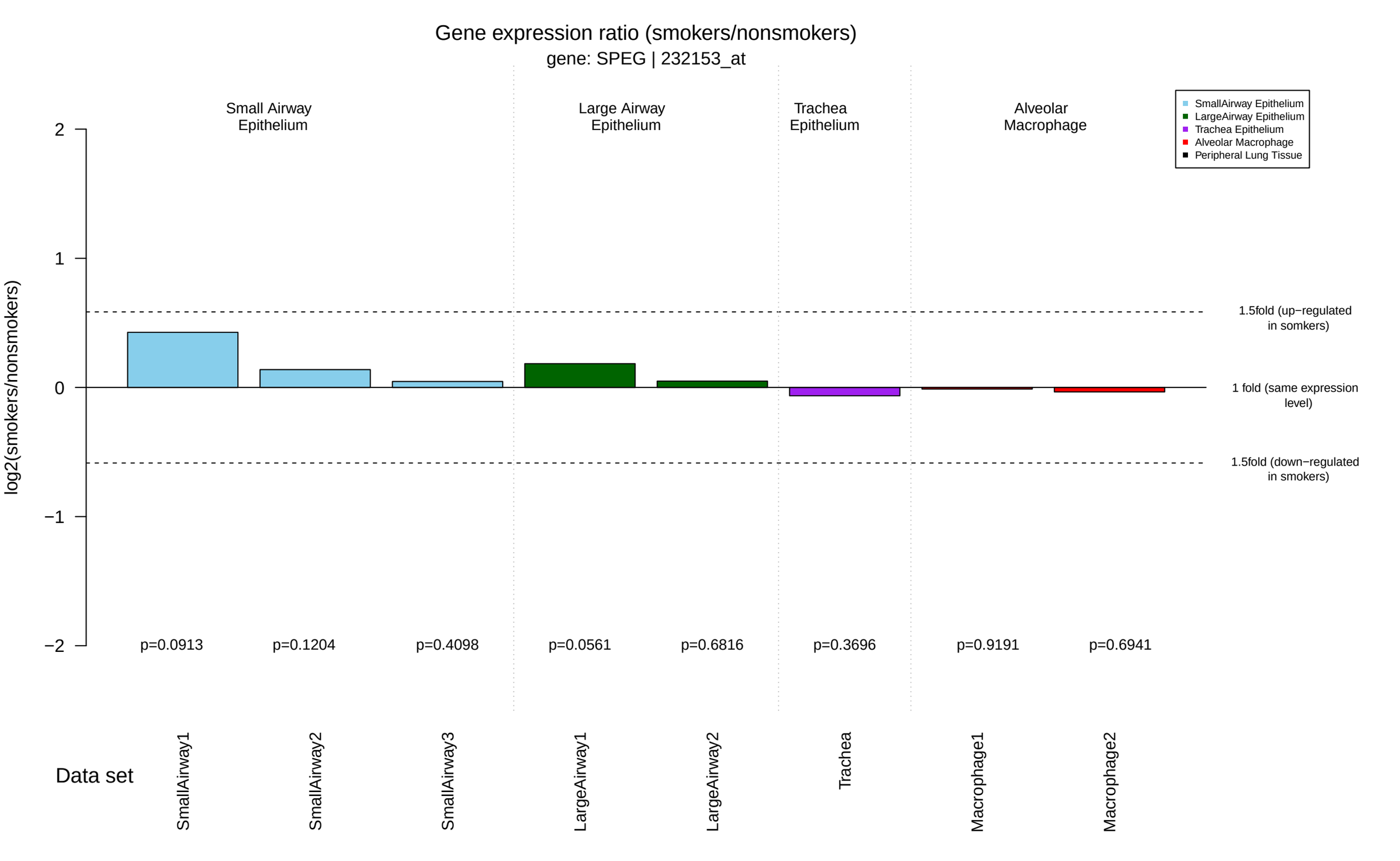
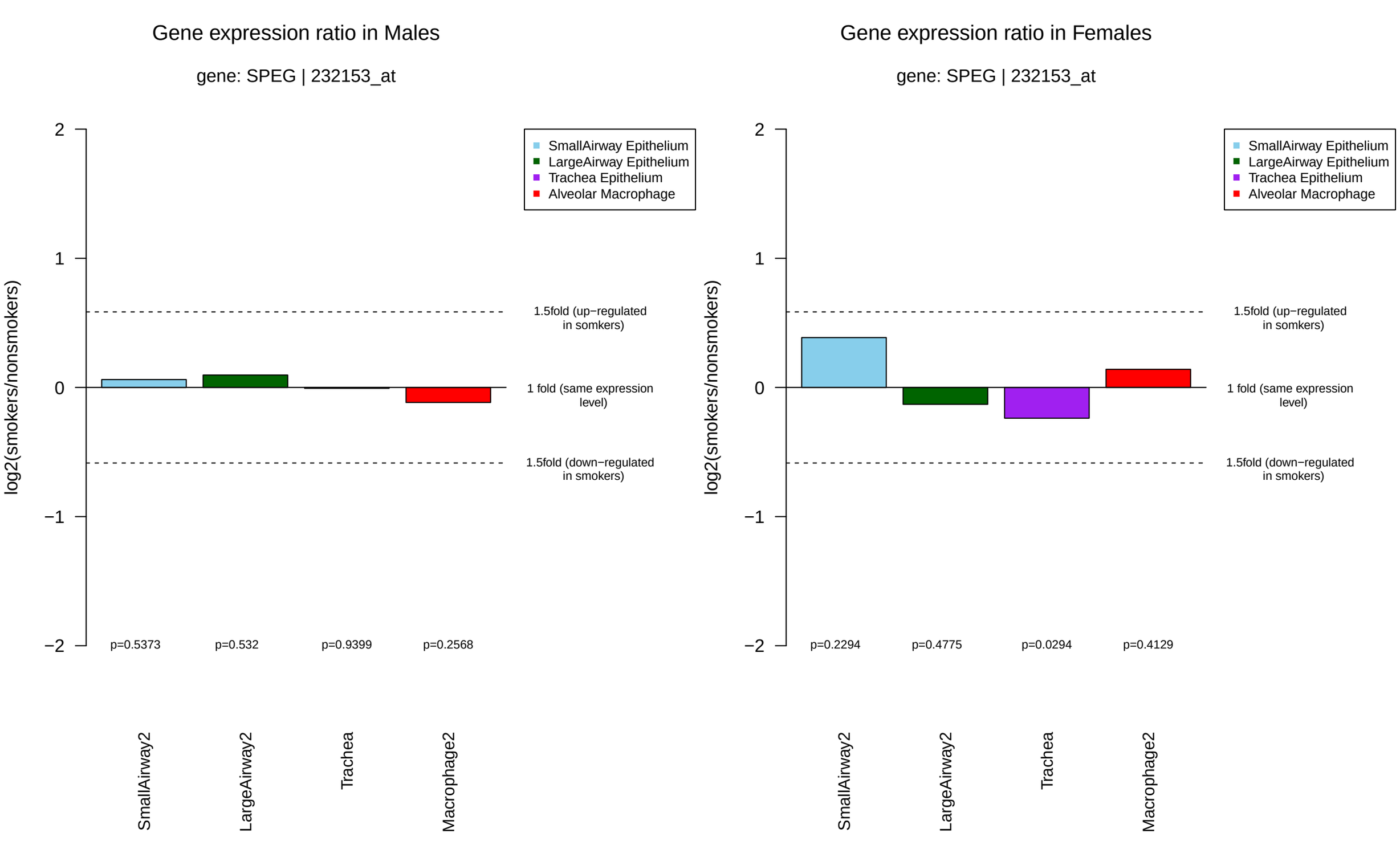
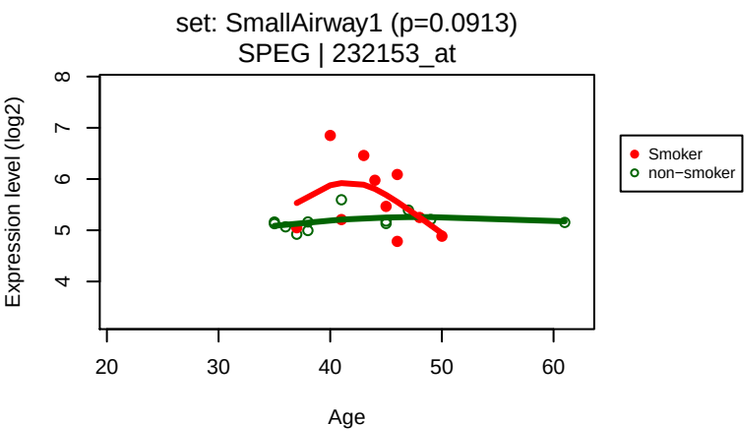
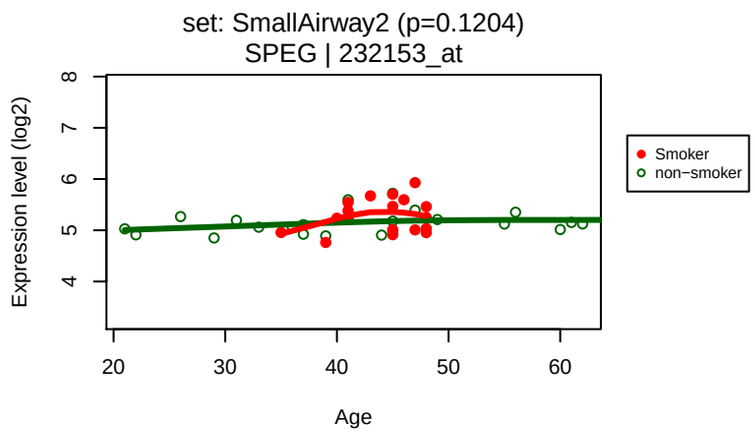
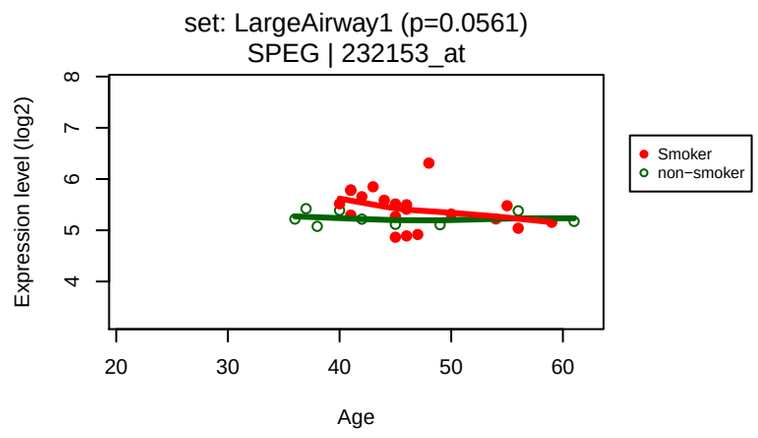

Gene expression of smokers and nonsmokers
gene: SPEG | 232153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0913
p=0.1204
p=0.4098
p=0.0561
p=0.6816
p=0.3696
p=0.9191
p=0.6941
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

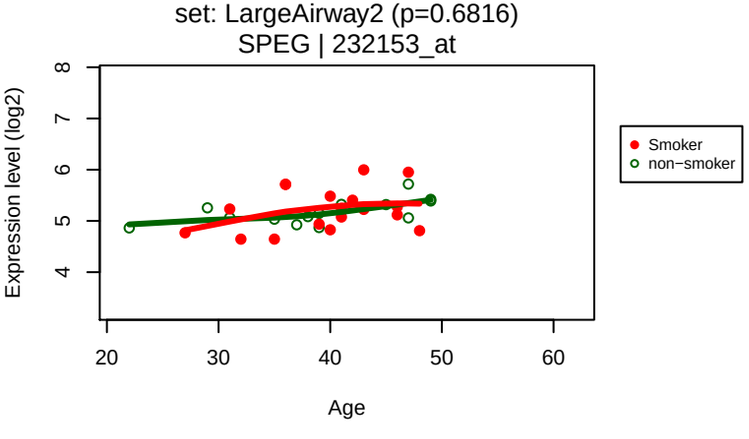

Gene expression ratio (smokers/nonsmokers)
gene: SPEG | 232153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0913
p=0.1204
p=0.4098
p=0.0561
p=0.6816
p=0.3696
p=0.9191
p=0.6941
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

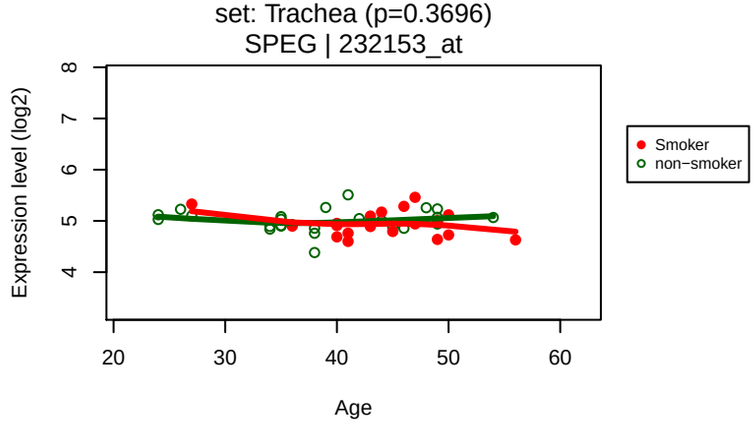

Gene expression ratio in Males
gene: SPEG | 232153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5373
p=0.532
p=0.9399
p=0.2568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPEG | 232153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2294
p=0.4775
p=0.0294
p=0.4129
SmallAirway2
LargeAirway2
Trachea
Macrophage2

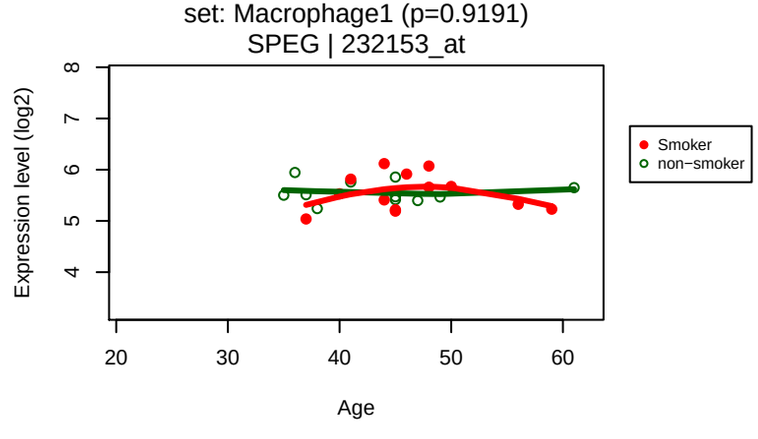

set: SmallAirway1 (p=0.0913)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker

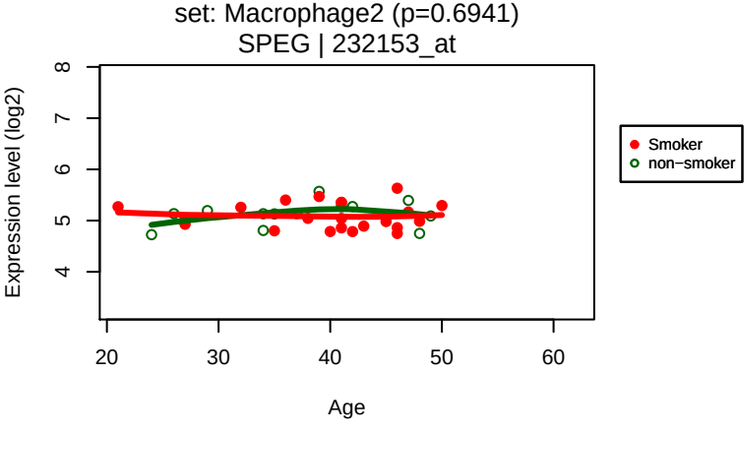

set: SmallAirway2 (p=0.1204)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker

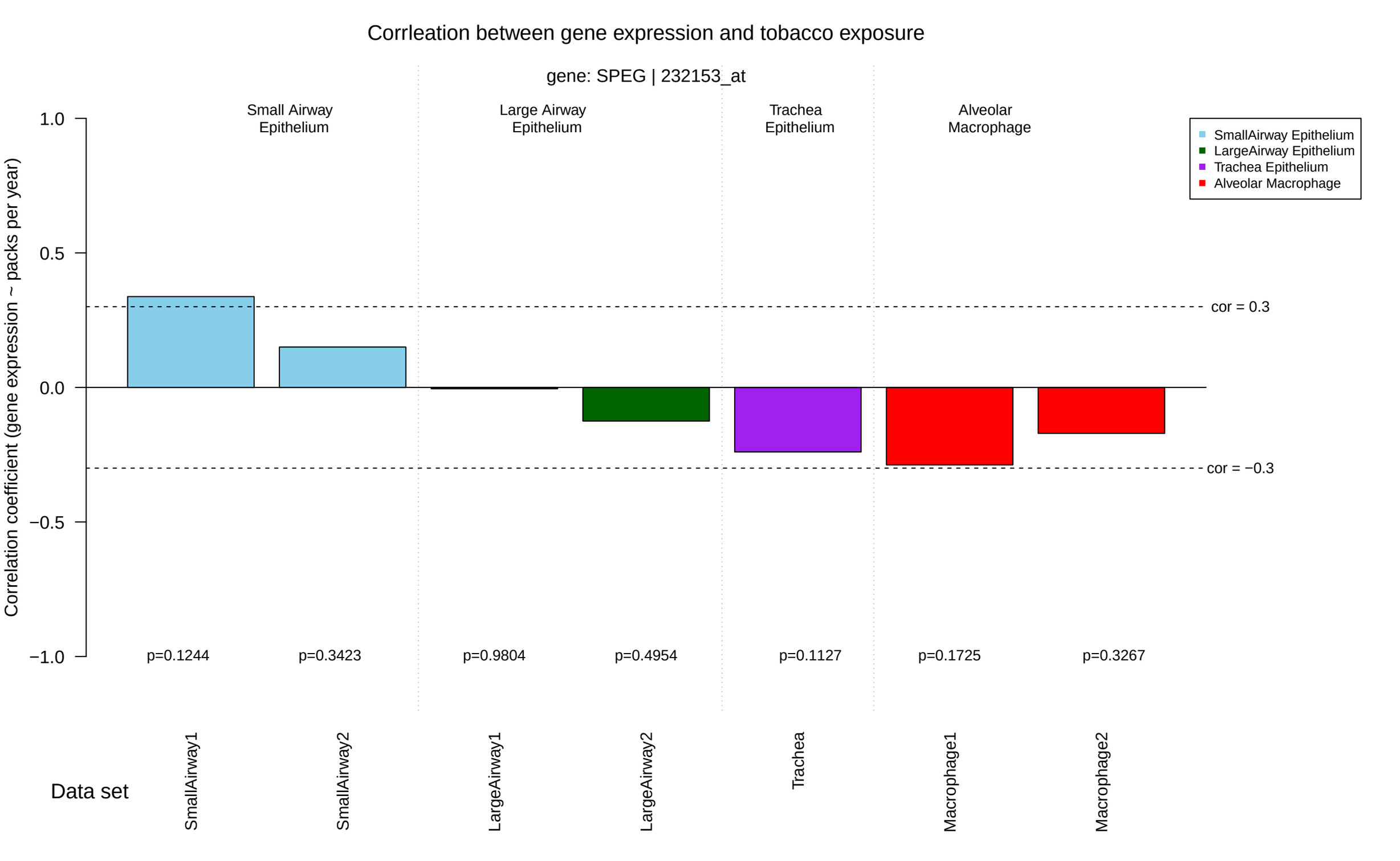

set: LargeAirway1 (p=0.0561)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6816)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3696)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9191)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6941)
SPEG | 232153_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPEG | 232153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1244
p=0.3423
p=0.9804
p=0.4954
p=0.1127
p=0.1725
p=0.3267
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2